

Quantum Computation and Information

Homework Set 2

1. Recall Pauli X, Y, Z matrices: $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

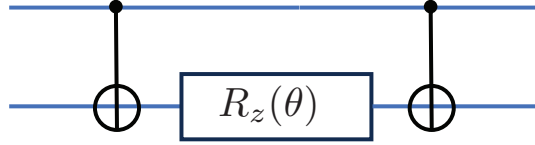
(a). Show that for any $j \in \{1, 2, 3\}$

$$e^{-i\theta\sigma_j} = I \cos \theta - i\sigma_j \sin \theta$$

Solutions: Note that $\sigma_j^2 = I$. So $\sigma_j^n = I$ if n is even and $\sigma_j^n = \sigma_j$ if n is odd. By Taylor expansion,

$$\begin{aligned} e^{-i\theta\sigma_j} &= \sum_{n=0}^{\infty} \frac{(-i\theta\sigma_j)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-i\theta\sigma_j)^{2n}}{(2n)!} + \sum_{m=0}^{\infty} \frac{(-i\theta\sigma_j)^{2m+1}}{(2m+1)!} \\ &= I \sum_{n=0}^{\infty} \frac{(-1)^n \theta^{2n}}{(2n)!} - i\sigma_j \sum_{m=0}^{\infty} \frac{(-1)^m \theta^{2m+1}}{(2m+1)!} = I \cos \theta - i\sigma_j \sin \theta. \end{aligned}$$

(b). Consider a system of two qubits. Denote by Z_0 and Z_1 the Pauli Z matrices on qubits 0 and 1, respectively. Show that the unitary transformation $U(\theta) = e^{-i\theta Z_0 Z_1}$ can be implemented by the following circuit with $R_z(\theta) = e^{-i\theta\sigma_3}$.



Solutions: Note that $U(\theta) |x_0 x_1\rangle = e^{-i(-1)^{x_0 \oplus x_1}} |x_0 x_1\rangle$ for $|x_0 x_1\rangle \in \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ (the basis states).

With input $|x_0 x_1\rangle$, the output of the above circuit is

$$\begin{aligned} |x_0 x_1\rangle &\Rightarrow |x_0(x_0 \oplus x_1)\rangle \Rightarrow e^{-i\theta(-1)^{x_0 \oplus x_1}} |x_0(x_0 \oplus x_1)\rangle \\ &\Rightarrow e^{-i\theta(-1)^{x_0 \oplus x_1}} |x_0((x_0 \oplus x_1) \oplus x_0)\rangle \\ &= e^{-i\theta(-1)^{x_0 \oplus x_1}} |x_0 x_1\rangle = U(\theta) |x_0 x_1\rangle. \end{aligned}$$

2. **(Amplitude Amplification)** Given a unitary operator U on a system of n qubits:

$$U |0^{\otimes n}\rangle = \cos \theta |\Psi_0\rangle + \sin \theta |\Psi_1\rangle,$$

where $\langle \Psi_0 | \Psi_0 \rangle = \langle \Psi_1 | \Psi_1 \rangle = 1$ and $\langle \Psi_0 | \Psi_1 \rangle = 0$. Define $S_1 = I - 2 |\Psi_1\rangle \langle \Psi_1|$, $S_i = I - 2 |0^{\otimes n}\rangle \langle 0^{\otimes n}|$, and $Q = -US_iU^{-1}S_1$.

(a). Show that

$$\begin{aligned} Q |\Psi_1\rangle &= \cos 2\theta |\Psi_1\rangle - \sin 2\theta |\Psi_0\rangle \\ Q |\Psi_0\rangle &= \sin 2\theta |\Psi_1\rangle + \cos 2\theta |\Psi_0\rangle \end{aligned}$$

The above relations implies the two-dimensional space spanned by $|\Psi_0\rangle$ and $|\Psi_1\rangle$ is invariant under operator Q .

Solutions: Note that $S_1 |\Psi_0\rangle = |\Psi_0\rangle$, $S_1 |\Psi_1\rangle = -|\Psi_1\rangle$, and

$$\begin{aligned} Q &= -US_iU^{-1}S_1 = -U(I - 2 |0^{\otimes n}\rangle \langle 0^{\otimes n}|)U^{-1}S_1 \\ &= (2U |0^{\otimes n}\rangle \langle 0^{\otimes n}|)U^{-1} - I)S_1 \\ &= (2\{\cos \theta |\Psi_0\rangle + \sin \theta |\Psi_1\rangle\}\{\cos \theta |\Psi_0\rangle + \sin \theta |\Psi_1\rangle\}^\dagger - I)S_1 \\ &= 2\{\cos \theta |\Psi_0\rangle + \sin \theta |\Psi_1\rangle\}\{\cos \theta |\Psi_0\rangle + \sin \theta |\Psi_1\rangle\}^\dagger (I - 2 |\Psi_1\rangle \langle \Psi_1|) - (I - 2 |\Psi_1\rangle \langle \Psi_1|) \\ &= 2\{\cos \theta |\Psi_0\rangle + \sin \theta |\Psi_1\rangle\}\{\cos \theta |\Psi_0\rangle - \sin \theta |\Psi_1\rangle\}^\dagger - (I - 2 |\Psi_1\rangle \langle \Psi_1|) \\ &= 2\{\cos \theta |\Psi_0\rangle + \sin \theta |\Psi_1\rangle\}\{\cos \theta |\Psi_0\rangle - \sin \theta |\Psi_1\rangle\}^\dagger - (I - 2 |\Psi_1\rangle \langle \Psi_1|). \end{aligned}$$

Thus,

$$\begin{aligned} Q |\Psi_1\rangle &= -2 \sin \theta \{\cos \theta |\Psi_0\rangle + \sin \theta |\Psi_1\rangle\} + |\Psi_1\rangle \\ &= (1 - 2 \sin^2 \theta) |\Psi_1\rangle - 2 \sin \theta \cos \theta |\Psi_0\rangle \\ &= \cos 2\theta |\Psi_1\rangle - \sin 2\theta |\Psi_0\rangle \end{aligned}$$

and

$$\begin{aligned} Q |\Psi_0\rangle &= 2 \cos \theta \{\cos \theta |\Psi_0\rangle + \sin \theta |\Psi_1\rangle\} - |\Psi_0\rangle \\ &= (2 \cos^2 \theta - 1) |\Psi_0\rangle + 2 \sin \theta \cos \theta |\Psi_1\rangle \\ &= \cos 2\theta |\Psi_0\rangle + \sin 2\theta |\Psi_1\rangle. \end{aligned}$$

(b). Show that

$$Q^k U |0^{\otimes n}\rangle = \cos (2k + 1)\theta |\Psi_0\rangle + \sin (2k + 1)\theta |\Psi_1\rangle, \quad k \in \mathbb{Z}_+.$$

Suppose θ is very small. If $|\Psi_1\rangle$ is the desired result, the above relation implies that the probability of obtaining the desired result increases quadratically with the number of times Q is applied.

Solutions: Proof by induction:

1) when $k = 0$, the above equation obviously holds.

2) suppose the above equation holds for $k = m$, i.e., $Q^m U |0^{\otimes n}\rangle = \cos (2m + 1)\theta |\Psi_0\rangle + \sin (2m + 1)\theta |\Psi_1\rangle$. Then for $k = m + 1$, we have

$$Q^k U |0^{\otimes n}\rangle = Q^{m+1} U |0^{\otimes n}\rangle = Q Q^m U |0^{\otimes n}\rangle$$

$$\begin{aligned}
&= Q(\cos(2m+1)\theta |\Psi_0\rangle + \sin(2m+1)\theta |\Psi_1\rangle) \\
&= \cos(2m+1)\theta Q|\Psi_0\rangle + \sin(2m+1)\theta Q|\Psi_1\rangle \\
&= \cos(2m+1)\theta \{\sin 2\theta |\Psi_1\rangle + \cos 2\theta |\Psi_0\rangle\} + \sin(2m+1)\theta \{\cos 2\theta |\Psi_1\rangle - \sin 2\theta |\Psi_0\rangle\} \\
&= \{\cos(2m+1)\theta \sin 2\theta + \sin(2m+1)\theta \cos 2\theta\} |\Psi_1\rangle \\
&\quad + \{\cos(2m+1)\theta \cos 2\theta - \sin(2m+1)\theta \sin 2\theta\} |\Psi_0\rangle \\
&= \sin(2m+3)\theta |\Psi_1\rangle + \cos(2m+3)\theta |\Psi_0\rangle \\
&= \sin(2k+1)\theta |\Psi_1\rangle + \cos(2k+1)\theta |\Psi_0\rangle,
\end{aligned}$$

i.e., the above equation holds for $k = m + 1$.

3. Consider the quantum Fourier transformation

$$QFT_{2^2} |x\rangle = \frac{1}{\sqrt{2^2}} \sum_{y=0}^{2^2-1} e^{i\frac{2\pi xy}{2^2}} |y\rangle = \frac{1}{\sqrt{2^2}} (|0\rangle + e^{i2\pi(.x_0)} |1\rangle)(|0\rangle + e^{i2\pi(.x_1x_0)} |1\rangle).$$

(a). Recall the notation introduced in lecture 3: $x \equiv x_1x_0$ and $y \equiv y_1y_0$. Show that

$$\frac{xy}{2^2} \pmod{1} \equiv y_1(.x_0) + y_0(.x_1x_0)$$

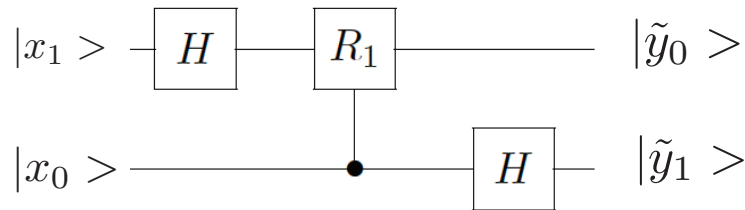
Solutions: Note that

$$\frac{xy}{2^2} = \frac{(2x_1 + x_0)(2y_1 + y_0)}{2^2} = x_1y_1 + (x_1y_0 + x_0y_1)2^{-1} + x_0y_02^{-2}.$$

Thus,

$$\begin{aligned}
\frac{xy}{2^2} \pmod{1} &= (x_1y_0 + x_0y_1)2^{-1} + x_0y_02^{-2} \pmod{1} \\
&= y_1\frac{x_0}{2} + y_0\left(\frac{x_1}{2} + \frac{x_0}{2^2}\right) \pmod{1} \\
&= y_1(.x_0) + y_0(.x_1x_0) \pmod{1}.
\end{aligned}$$

(b). Show that QFT_{2^2} can be computed by the following circuit:



where $|\tilde{y}_0\rangle = (|0\rangle + e^{i2\pi(.x_1x_0)} |1\rangle)/\sqrt{2}$, $|\tilde{y}_1\rangle = (|0\rangle + e^{i2\pi(.x_0)})/\sqrt{2}$, and $R_1 = \begin{pmatrix} 1 & 0 \\ 0 & e^{\frac{i\pi}{2}} \end{pmatrix}$.

Solutions:

$$\begin{aligned}
|x_1 x_0\rangle &\Rightarrow \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_1} |1\rangle) |x_0\rangle \Rightarrow \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{x_1} e^{\frac{i\pi}{2} x_0} |1\rangle) |x_0\rangle \\
&\Rightarrow \frac{1}{\sqrt{2}^2}(|0\rangle + (-1)^{x_1} e^{\frac{i\pi}{2} x_0} |1\rangle)(|0\rangle + (-1)^{x_0} |1\rangle) \\
&= \frac{1}{\sqrt{2}^2}(|0\rangle + e^{\frac{i2\pi x_1}{2}} e^{\frac{i2\pi}{2^2} x_0} |1\rangle)(|0\rangle + e^{\frac{i2\pi x_0}{2}} |1\rangle) \\
&= \frac{1}{\sqrt{2}^2}(|0\rangle + e^{i2\pi(.x_1 x_0)} |1\rangle)(|0\rangle + e^{i2\pi(.x_0)} |1\rangle)
\end{aligned}$$